Automated Stroke Prediction Using Machine Learning: An Explainable and Exploratory Study with a Web Application for Early Intervention

Abstract

Stroke is a dangerous medical disorder that occurs when blood flow to the brain is disrupted, resulting in neurological impairment. It is a big worldwide threat with serious health and economic implications. To solve this, researchers are developing automated stroke prediction algorithms, which would allow for early intervention and perhaps save lives. The number of people at risk for stroke is growing as the population ages, making precise and effective prediction systems increasingly critical. The goal of this study was study aimed to achieve three objectives: (i) to create a trustworthy machine learning model to predict stroke disease, (ii) to address the severe class imbalance issue that results from the stroke patients' class being substantially smaller than the healthy class, and (iii) by using Mutual Information Score, Chi-Square Score, and ANOVA test, find the import feature(iv) to interpret the model output to gain a better comprehension of the decision-making process (v) balancing the dataset from the ratio of 19: 1 for No Stroke: Stroke to equal ratio using SMOTE Analysis (vi) Propose an End-to-End smart healthcare system through an android application. In a comparison examination with six well-known classifiers, the effectiveness of the proposed ML technique was explored in terms of metrics relating to both generalization capability and prediction accuracy. To give insight into the black-box machine learning models, we also studied two kinds of explainable techniques, namely SHAP and LIME, in this study. SHAP (Shapley Additive Explanations) and LIME (Local Interpretable Model-agnostic Explanations) are well-established and reliable approaches for explaining model decision-making, particularly in the medical industry. The findings of the experiment revealed that more complicated models outperformed simpler ones, with the top model obtaining almost 91% accuracy and the other models achieving 83-91% accuracy. The proposed framework, which includes global and local explainable methodologies, can aid in standardizing complicated models and gaining insight into their decision-making, which can enhance stroke care and treatment.

Existing System

In Existing Stroke often causes due to blood flow stop to brain and this is one of the deadly diseases. Patient life can be saved and stroke can be avoided by timely and accurate detection.

Existing detection technique requires heavy resources and they make time for prediction. To overcome from this problem many machine learning algorithms were introduced as they are very accurate in medical diseases prediction but existing techniques were suffering from data leakage such as improper handling or missing values, improper categorical data calculation etc. No existing techniques were employing any Explainable model (XAI) which can show which features are helping most in detecting stroke so doctor can give priority on such features for faster recovery. These explainable features can be Smoking, Age, BMI and may be other features.

DISADVANTAGE:

- Accuracy Less
- More Time Taking

PROPOSED SYSTEM

In propose work author is employing different processing techniques such as Removing missing values, Imbalance data handling using SMOTE and relevant features selection using CHI2 algorithm. All these processed features will get trained on 6 different algorithms such as Random Forest, KNN, SVM, Logistic Regression, XGBOOST and Naive Bayes. In all algorithm Random Forest is giving high accuracy and each algorithm performance is evaluated in terms of accuracy, precision, recall and FSCORE.

ADVANTAGE:

- Accuracy more.
- Less Time Taking.

<u>SYSTEM REQUIREMENTS</u>

HARDWARE REQUIREMENTS:

Processor	-	Intel i3 or higher
Speed	-	1.1 GHz
RAM	-	4 GB (min)

Hard Disk	-	500 GB (min)

SOFTWARE REQUIREMENTS:

Operating System	-	Windows 10 or above
Programming Language	-	Python (3.7.0) with Jupiter notebook